Experiment no:8

Date:

Experiment Title: Testing of Requirements

Aim:

The aim of this experiment is to analyze and test the given system requirements to ensure that they are clear, complete, consistent, and suitable for implementation. It also aims to verify whether the requirements can be effectively used to design accurate test cases and support system development.

Condition:

The given requirements must satisfy certain conditions such as clarity, correctness, completeness, and consistency. They should be free from ambiguity and must be easily understandable. The requirements should also be testable so that their correctness can be verified using appropriate testing techniques without any confusion.

Description:

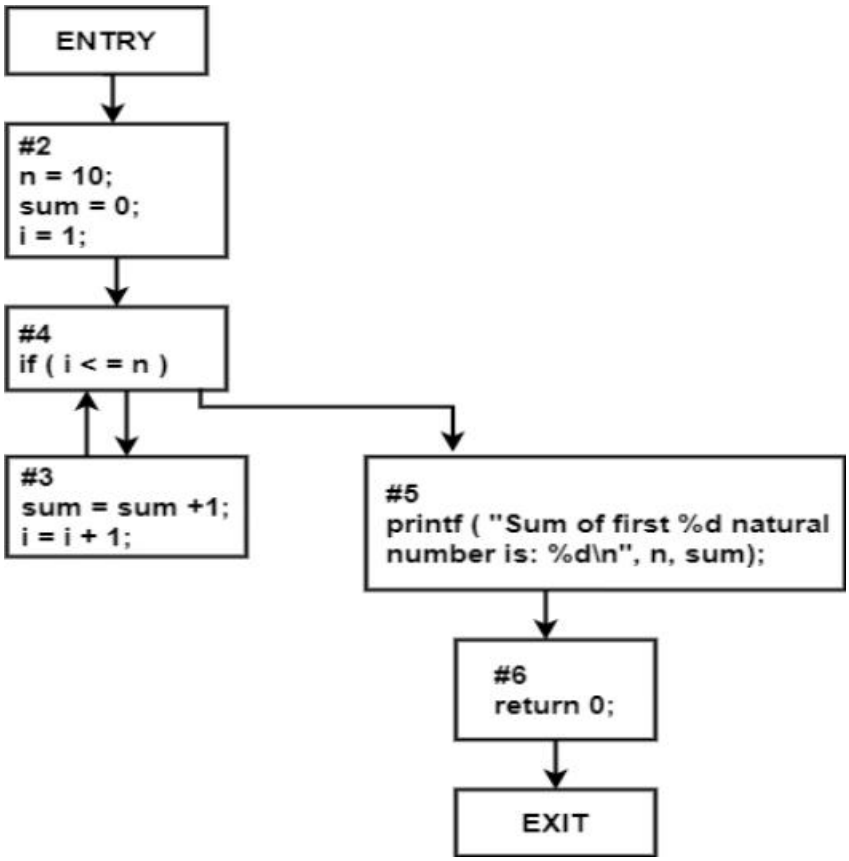
Requirement testing is a process in software engineering that focuses on validating and verifying the system requirements before implementation. It ensures that the requirements accurately represent the user needs and system functionality. This process helps in identifying errors, missing information, or inconsistencies in the requirements at an early stage.

Methodology:

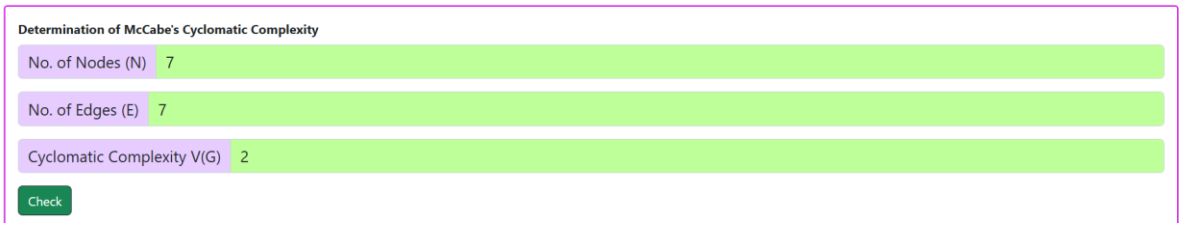
The methodology adopted in this experiment involves analyzing the requirements and applying testing techniques such as requirement inspection, review, and validation. Each requirement is examined carefully, and test cases are designed based on the expected input and output conditions. This structured approach ensures that all requirements are properly evaluated.

Exercise 1:

Control Flow Graph of the program

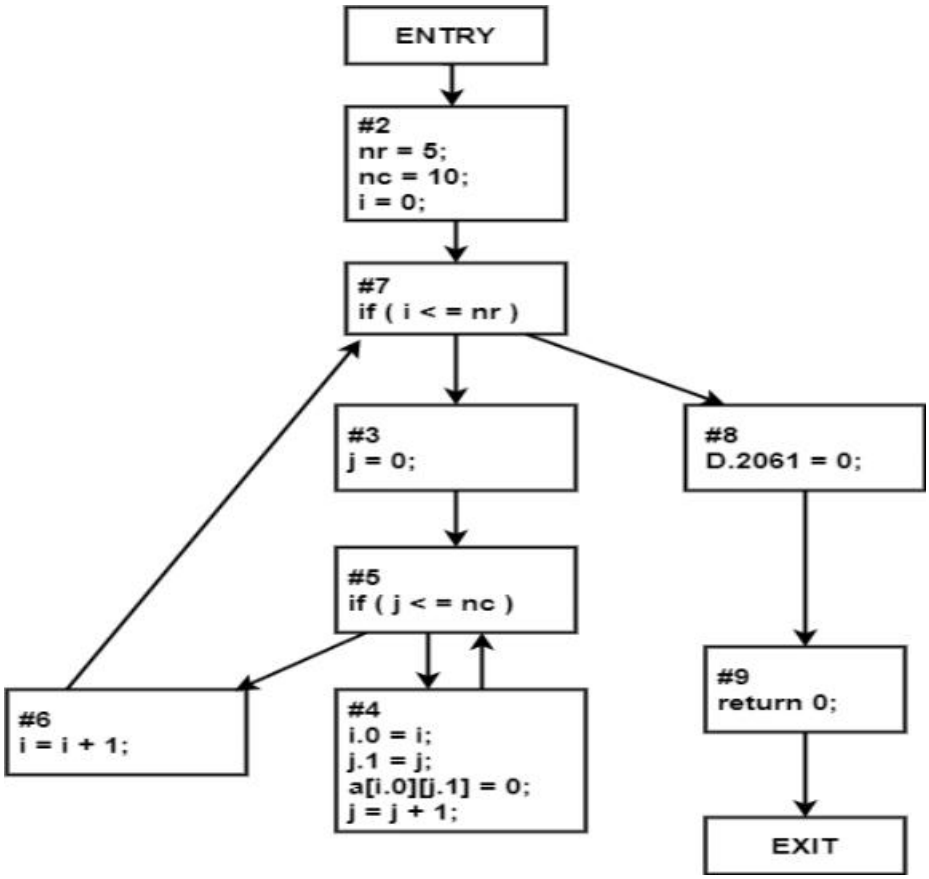


OUTPUT:



Exercise 2:

Control Flow Graph of the program



Output:

Determination of McCabe's cyclomatic complexity

No. of Nodes (N)	10
No. of Edges (E)	11
Cyclomatic Complexity V(G)	3

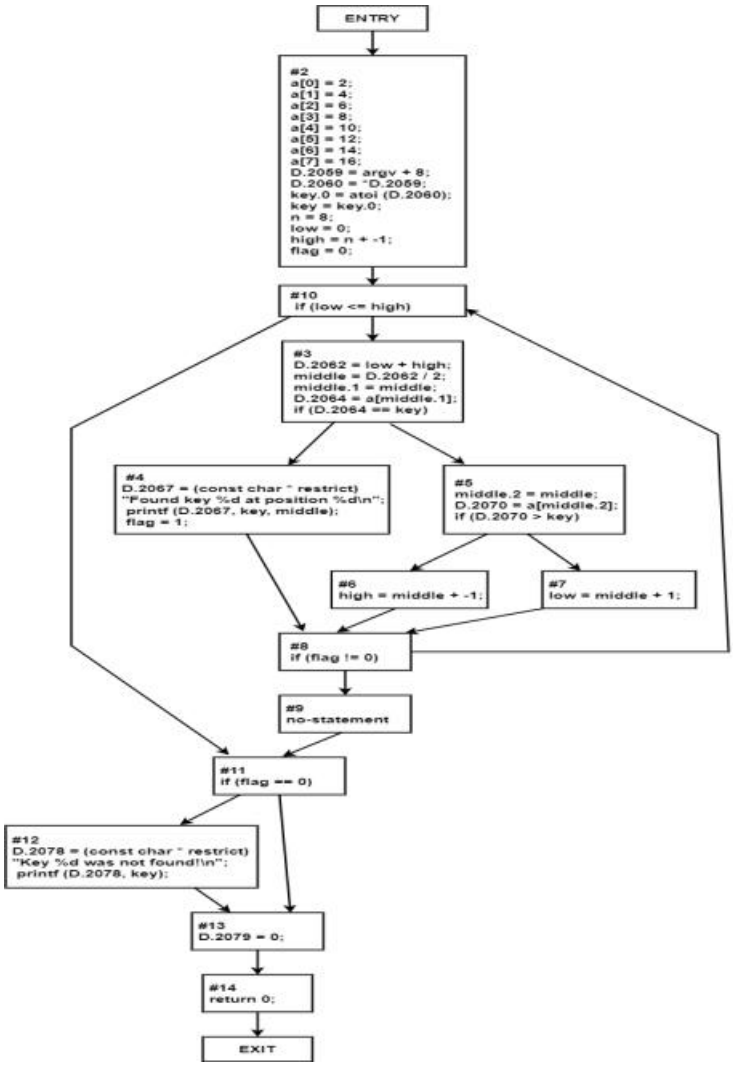
Check

The linearly independent paths for the above CFG are:

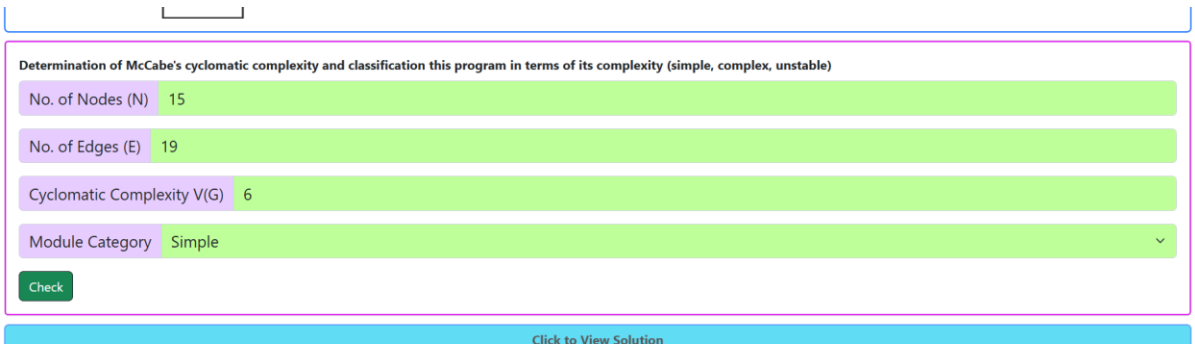
Path 1	2	-	7	-	8	-	9	Check												
Path 2	2	-	7	-	3	-	5	-	6	-	7	-	8	-	9	Check				
Path 3	2	-	7	-	3	-	5	-	4	-	5	-	6	-	7	-	8	-	9	Check

Exercise 3:

Control Flow Graph of the program



Output:



Working

The working of this experiment begins with understanding the given requirements and identifying key functionalities. Based on these requirements, test cases are created to verify their correctness. Each requirement is tested by comparing expected results with actual outcomes. Any discrepancies found during testing are noted and corrected to improve the quality of requirements.

Result

The result of the experiment shows that the requirements were successfully tested and validated. The testing process helped in identifying ambiguities and ensuring that all requirements are clear, consistent, and testable. This confirms that the requirements are suitable for further stages of software development.

Conclusion

Thus, the experiment on testing of requirements was successfully completed. It highlights the importance of validating requirements at an early stage to reduce errors and improve system quality. Proper requirement testing ensures a strong foundation for the development of reliable software systems.